

Prajwal Khanal

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Faculty of Geo-Information Science and Earth Observation (ITC), University of Twente, Enschede

INTERESTS	Remote sensing of vegetation; soil–vegetation–atmosphere interaction; land surface models and deep learning.
EDUCATION	PhD, Geo-Information Science and Earth Observation Feb 2023—... <i>University of Twente and Planet Labs, the Netherlands</i> – WUNDER: Prediction of field-scale soil moisture, water and carbon fluxes for vegetation monitoring in the Netherlands. – Supervisors: <u>Prof. dr. Bob Su</u>, <u>dr. Yijian Zeng</u> and <u>dr. Joost Beckers</u>. M.Sc., Environmental Engineering, Technical University of Munich Nov 2020—Dec 2022 – Graduated 1.4/5.0 (1 = best). Thesis: <u>Role of surface and sub-surface soil moisture for vegetation functioning</u>. B.E., Civil Engineering, Institute of Engineering, Tribhuvan University, Nepal Nov 2014—Aug 2018 – Ranked 20th out of 192, 77.7%. Thesis: Flood inundation mapping of Babai River.
EMPLOYMENT	Research Intern, Max Planck Institute for Biogeochemistry, Jena Mar 2022—Feb 2023 – Published a study on <u>near-surface soil moisture vs. terrestrial water storage control on vegetation globally</u>. Student Research Assistant, Technical University of Munich Nov 2021—Feb 2023 – Prepared flood inundation maps for the Ouémé (Benin) and Naryn (Kyrgyzstan) rivers using HEC-RAS. Working Student, Jena-Geos Ingenieurbüro GmbH, Munich May 2021—Dec 2021 – Developed an Excel/VBA greenhouse-gas accounting tool for small-scale enterprises. Freelance Civil Engineer, Executive Consulting Engineering and Planner (ECEP) Apr 2020—Dec 2020 <i>Client: Department of Water Supply and Sanitation, Government of Nepal.</i> – Proposed a multi-source redesign (treated river and spring water) with decentralised cluster tanks, and drafted the business plan for the Khalanga Water Supply Project (commissioned 2020), serving about 1,500 households.
PUBLICATIONS	<i>Peer-reviewed</i> Moutahir, H., Sulzer, M., Kiese, R., ... Khanal, P. , et al. (2026). <u>Matching scales of eddy covariance measurements and process-based modelling: assessing spatiotemporal dynamics of carbon and water fluxes in a mixed forest in Southern Germany</u> . <i>Biogeosciences</i> , 23, 1719–1738. Zeng, Y., Alidoost, F., Schilperoort, B., ... Khanal, P. , et al. (2025). <u>Towards an open soil-plant digital twin based on the STEMMUS-SCOPE model, following open science</u> . <i>Computers & Geosciences</i> , 205, 106013. Khanal, P. , Hoek van Dijke, A. J., Schaffhauser, T., et al. (2024). <u>Relevance of near-surface soil moisture vs. terrestrial water storage for global vegetation functioning</u> . <i>Biogeosciences</i> , 21(6), 1533–1547. Bhattarai, P., Khanal, P. , Tiwari, P., et al. (2019). <u>Flood inundation mapping of Babai basin using HEC-RAS and GIS</u> . <i>Journal of the Institute of Engineering</i> , 15(2), 32–44. <i>Under review</i> Khanal, P. , Zeng, Y., Daoud, M. G., Beckers, J., Su, Z. <u>Investigating the impact of multilayer resistance schemes on simulating land surface temperature and turbulent fluxes: evaluation within the STEMMUS-SCOPE model</u> . Khanal, P. , et al. <u>Deep learning soil moisture and fluxes with multimodal foundation-model embeddings</u> . <i>Conference contributions</i> Khanal, P. , et al. (2026). Bridging the scale gap: daily 10-metre soil moisture, evapotranspiration and carbon fluxes via multimodal deep learning. <i>Workshop on Machine Learning for Land and Hydrology</i> , ECMWF, Reading, UK. Khanal, P. , Hoek van Dijke, A., Zeng, Y., Orth, R. (2023). <u>Near-surface vs. sub-surface soil moisture impacts on vegetation functioning</u> . <i>EGU General Assembly</i> , Vienna, EGU23-7670.
AWARDS	Second place, Hackathon on Machine Learning for Earth System Modelling Aug 2026 – Quantization technique in geo foundation model. CESOC, ECMWF, TRA Modelling, FZ Jülich, Cologne.
SCHOLARSHIPS	Deutschlandstipendium, merit scholarship, Technical University of Munich 2021—2022 Merit-based scholarship, Tribhuvan University ; 8th of 9048 in the entrance exam 2014—2018
TEACHING	Teaching Assistant, Technical University of Munich 2021—2022 – River Basin Modelling; Integrated Water Resources Management; Rapidly Varying Flow. Assistant Lecturer, Cosmos College of Management and Technology, Nepal Apr 2020—Dec 2020 – Tutored Fluid Mechanics and Hydraulics; designed and taught a Python and GIS course for water resources.
VOLUNTEERING	Member, Young Hydrologic Society Netherlands (YHS-NL) Nov 2023—... – Organised the YHS-NL Symposium 2025 at ITC: 10 speakers from six Dutch universities. President, Nepal Students' Union, Pulchowk Campus Unit Dec 2017—Nov 2018 – Organised a national civil engineering exhibition in 2018, with about 10,000 visitors.
SOFTWARE	WUNDER data viewer, ITC, University of Twente 2026 – Created a small dashboard for the food forest farmers to monitor their fields. <u>Source</u>.
SKILLS	<i>Modelling</i> : HEC-RAS, SWAT, HEC-HMS, GIS · <i>Programming</i> : Python, R, Google Earth Engine, HPC (SLURM) <i>Languages</i> : English (C2), German (A1), Dutch (A2), Hindi, Nepali